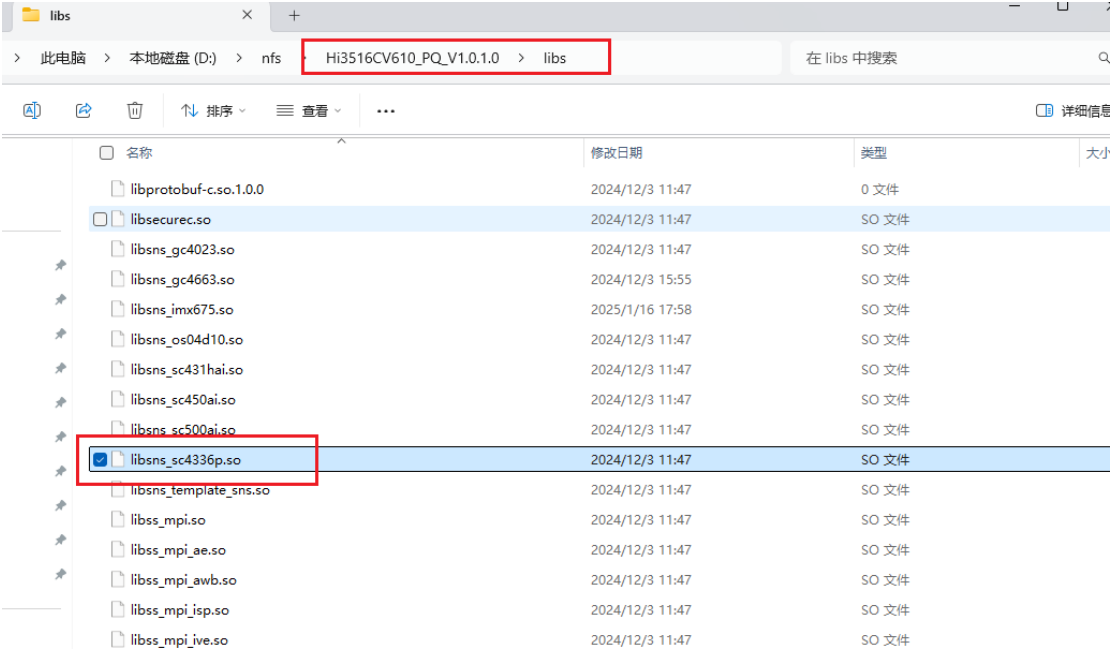
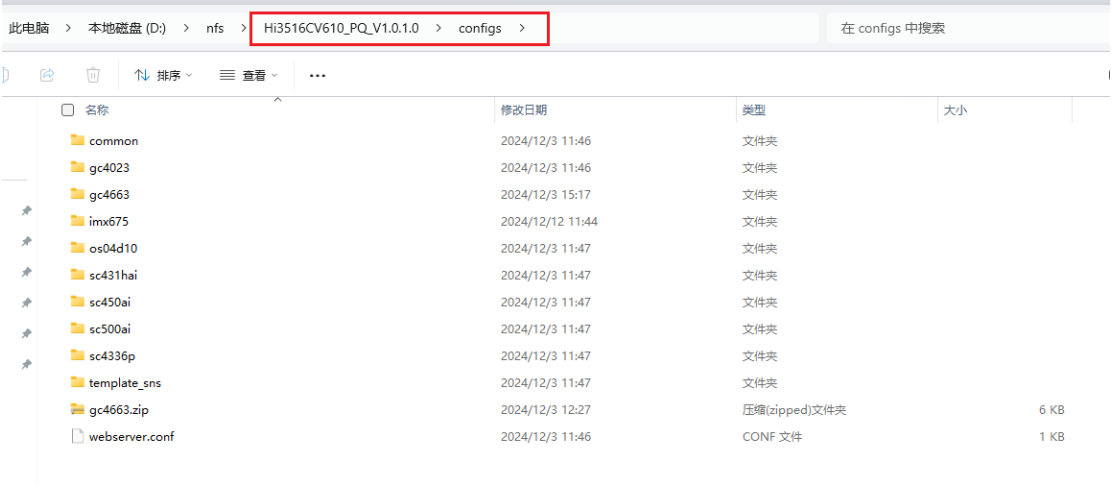


1 目录、参数、配置文件解释

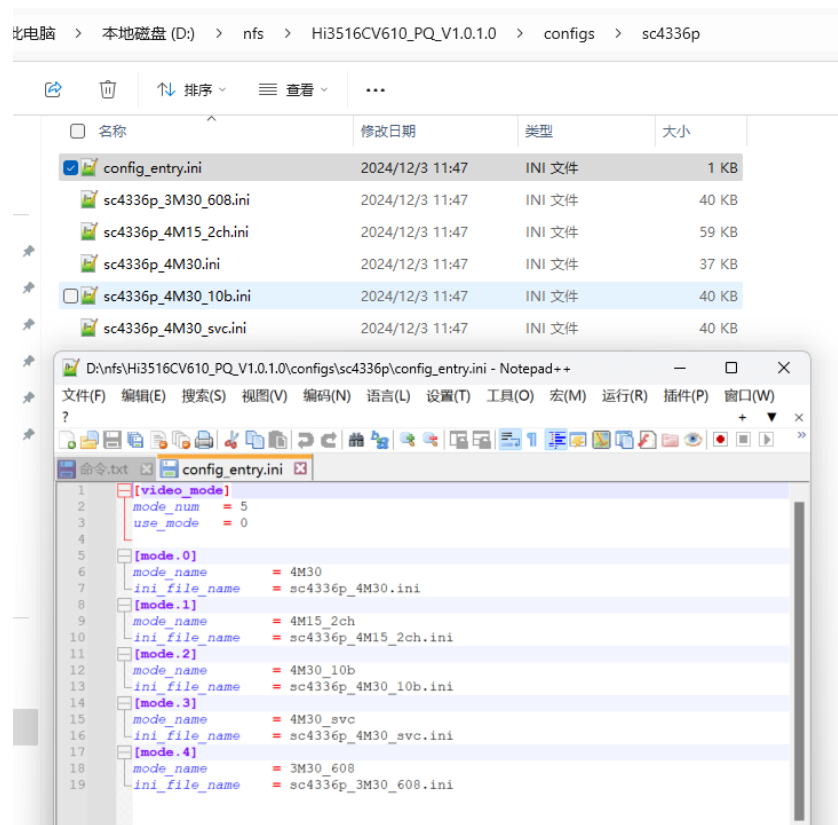
1.1 目录说明

(1) 运行板端pqtools.sh的脚本，需要依赖配置文件，配置文件的路径在pqtools的config目录下。每个文件夹中表示一款sensor的ini配置文件。

(2) 将sensor驱动编译之后，生成so文件，将sensor驱动库文件放入libs目录下。



1.2 参数说明



以sc4336p为例子，运行板端pqtools脚本的命令：

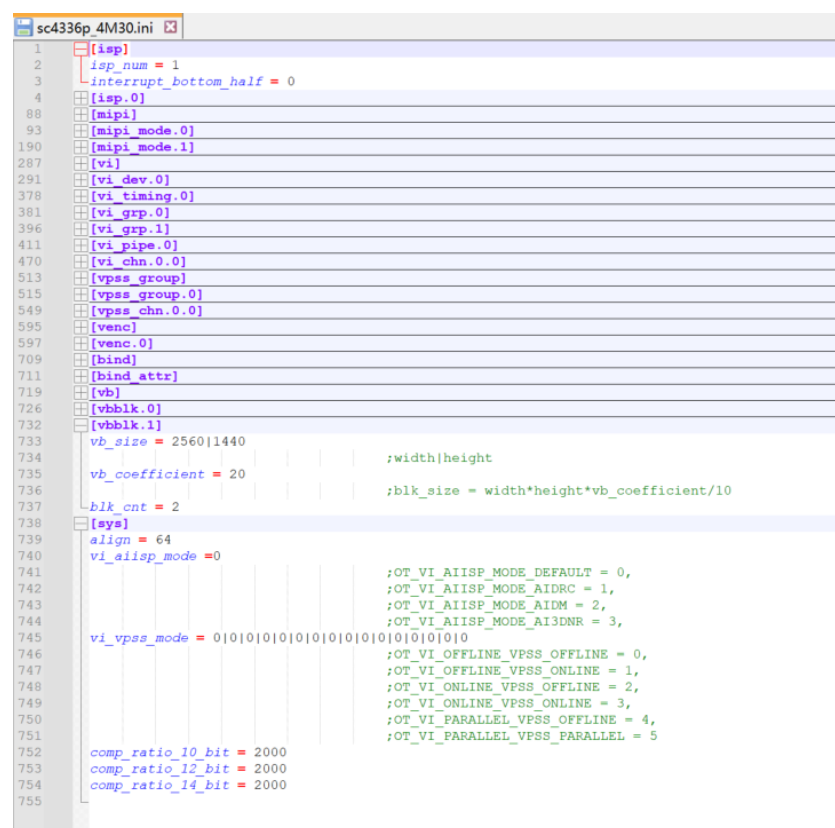
如执行 **./PQTools.sh -a sc4336p 0** 启动板端pqtools

参数sc4336p：表示从configs\sc4336p文件夹下找到config_entry.ini

参数0：表示从config_entry.ini文件中读取模式0，也就是加载

sc4336p_4M30.ini这个配置文件启动pqtools.sh脚本

1.3 ini 配置文件解释



```
1 [isp]
2   isp_num = 1
3   interrupt_bottom_half = 0
4   [isp.0]
88   [mipi]
93   [mipi_mode.0]
190  [mipi_mode.1]
287  [vi]
291  [vi.dev.0]
378  [vi.timing.0]
381  [vi.grp.0]
396  [vi.grp.1]
411  [vi.pipe.0]
470  [vi.chn.0.0]
513  [vpss_group]
515  [vpss_group.0]
549  [vpss_chn.0.0]
595  [venc]
597  [venc.0]
709  [bind]
711  [bind.attr]
719  [vb]
726  [vbblk.0]
732  [vbblk.1]
733  vb_size = 2560|1440
734                                     ;width|height
735  vb_coefficient = 20
736                                     ;blk_size = width*height*vb_coefficient/10
737  blk_cnt = 2
738  [sys]
739  align = 64
740  vi_aiisp_mode = 0
741                                     ;OT_VI_AIISP_MODE_DEFAULT = 0,
742                                     ;OT_VI_AIISP_MODE_AIDRC = 1,
743                                     ;OT_VI_AIISP_MODE_AIDM = 2,
744                                     ;OT_VI_AIISP_MODE_AI3DNR = 3,
745  vi_vpss_mode = 0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0
746                                     ;OT_VI_OFFLINE_VPSS_OFFLINE = 0,
747                                     ;OT_VI_OFFLINE_VPSS_ONLINE = 1,
748                                     ;OT_VI_ONLINE_VPSS_OFFLINE = 2,
749                                     ;OT_VI_ONLINE_VPSS_ONLINE = 3,
750                                     ;OT_VI_PARALLEL_VPSS_OFFLINE = 4,
751                                     ;OT_VI_PARALLEL_VPSS_PARALLEL = 5
752  comp_ratio_10_bit = 2000
753  comp_ratio_12_bit = 2000
754  comp_ratio_14_bit = 2000
755
```

上面打开了sc4336p的ini配置文件，我们将其折叠，可以看到isp-mipi-vi-vpss-venc-vb-sys相关的参数的配置，这些参数配置，其实和sample vio代码中的配置参数是有一一对应的关系的。启动pqtools.sh脚本也就相当于是用这些参数配置启动业务。

适配一款新的sensor，一般需要在ini中修改

- (1) sensor_type
- (2) sensor so库地址路径
- (3) i2c_dev : i2c设备地址
- (4) mipi lane_id: 接的几lane
- (5) 分辨率（直接全局替换所有的分辨率参数）
- (6) vb数量（可能需要调整）
- (7) vi vpss模式（建议使用离线模式跑，离线模式下vi能显示帧率）

2 新 sensor 适配 ini 流程

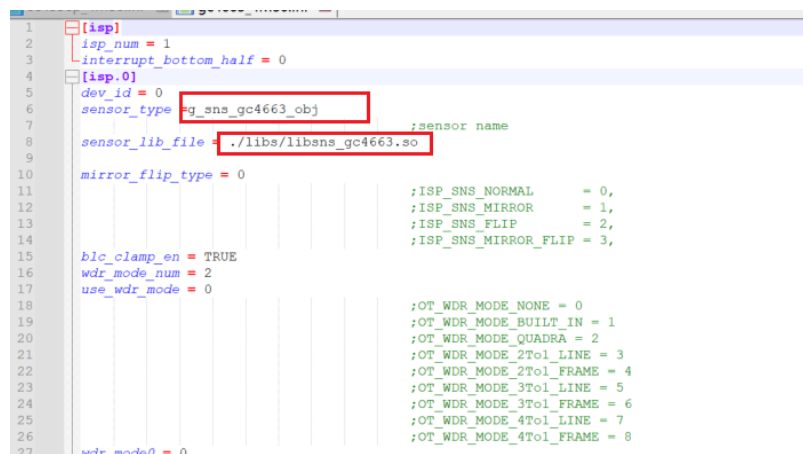
新来一款sensor，例如适配gc4663，改如何适配pqtools.sh脚本的配置文件

呢？

- (1) 将编译的sensor驱动so库放入\Hi3516CV610_PQ_V1.0.1.0\libs目录下。
- (2) 在config文件夹下创建gc4663文件夹，在文件夹中创建config_entry.ini
创建gc4663_4M30.ini，这个文件是直接从sc4336p的sc4336p_4M30.ini文件复制过来，修改一下名字，是在已有的ini文件的基础上进行修改。



- (3) 修改 sensor_type
- (4) sensor so库地址路径



- (5) 修改sensor i2c设备号如有修改，需要修改此处

```
sc4336p_4M30.ini  gc4663_4M30.ini
4 [isp.0]
5 dev_id = 0
6 sensor_type = g_sns_gc4663_obj
7 ;sensor name
8 sensor_lib_file = ./libs/libsns_gc4663.so
9
10 mirror_flip_type = 0
11
12 ;ISP_SNS_NORMAL = 0,
13 ;ISP_SNS_MIRROR = 1,
14 ;ISP_SNS_FLIP = 2,
15 ;ISP_SNS_MIRROR_FLIP = 3,
16
17 b1c_clamp_en = TRUE
18 wdr_mode_num = 2
19 use_wdr_mode = 0
20
21 ;OT_WDR_MODE_NONE = 0
22 ;OT_WDR_MODE_BUILT_IN = 1
23 ;OT_WDR_MODE_QUADRA = 2
24 ;OT_WDR_MODE_2To1_LINE = 3
25 ;OT_WDR_MODE_2To1_FRAME = 4
26 ;OT_WDR_MODE_3To1_LINE = 5
27 ;OT_WDR_MODE_3To1_FRAME = 6
28 ;OT_WDR_MODE_4To1_LINE = 7
29 ;OT_WDR_MODE_4To1_FRAME = 8
30
31 wdr_mode0 = 0
32 use_mipi_mode0 = 0
33 ;wdr_mode0 use mipi_mode.0
34
35 wdr_mode1 = 3
36 use_mipi_mode1 = 1
37 ;wdr_mode1 use mipi_mode.1
38
39 wnd_rect.x = 0
40 wnd_rect.y = 0
41 wnd_rect.width = 2560
42 wnd_rect.height = 1440
43 sns_size.width = 2560
44 sns_size.height = 1440
45 frame_rate = 30
46 bayer_format = 1
47
48 ;OT_ISP_BAYER_FGGB=0
49 ;OT_ISP_BAYER_GRGB=1
50 ;OT_ISP_BAYER_GBRG=2
51 ;OT_ISP_BAYER_BGGR=3
52
53 sns_mode = 0
54 sns_flip_en = FALSE
55 sns_mirror_en = FALSE
56 mipi_crop_en = FALSE
57 mipi_crop_offset.x = 0
58 mipi_crop_offset.y = 0
59 mipi_crop_offset.width = 2560
60 mipi_crop_offset.height = 1440
61 ;ot_isp_sns_commbus
62 sns_type = 0
63
64 ;OT_ISP_SNS_TYPE_I2C = 0
65 ;OT_ISP_SNS_TYPE_SSP = 1
66 ;OT_ISP_SNS_TYPE_BUTT = 2 /*do not set anything, use default setting*/
67
68 i2c_dev = 0
69 bit4_ssp_dev = -1
70 bit4_ssp_cs = -1
71 ;ot_isp_ctrl_param
72 be_buf_num = 8
73 proc_param = 30
74 stat_interval = 1
75 update_pos = 0
76 interrupt_time_out = 200
77 pwm_num = 7
78 port_interrupt_delay = 0
79 ldoi_tpr_fit_en = FALSE
80 ob_stats_update_pos = 0
81
82 ;OT_ISP_UPDATE_OB_STATS_FE_FRAME_END = 0 /* delay 1 frame */
83 ;OT_ISP_UPDATE_OB_STATS_FE_FRAME_START = 1 /* delay 2 frame */
84
85 alg_run_select = 0
86
87 ;OT_ISP_ALG_RUN_NORM = 0
88 ;OT_ISP_ALG_RUN_FE_ONLY = 1
89
90 isp_run_wakeup_select = 0
91
92 ;OT_ISP_RUN_WAKEUP_FE_START = 0
93 ;OT_ISP_RUN_WAKEUP_BE_END = 1
94
95 quick_start_en = FALSE
```

(6) 修改lane_id mipi_mode.0 和mipi_mode.1中建议都修改了
可以根据实际接入的lane进行配置

4lane: mipi_attr.lane_id = 0|1|2|3|-1|-1|-1|-1|-1

2lane: mipi_attr.lane_id = 0|2|-1|-1|-1|-1|-1|-1|-1

2lane: mipi_attr.lane_id = 1|3|-1|-1|-1|-1|-1|-1|-1

1lane: mipi_attr.lane_id = 0|-1|-1|-1|-1|-1|-1|-1|-1

如果以4lane的连接接入，初始化序列为2lane

mipi_attr.lane_id = 0|1|-1|-1|-1|-1|-1|-1|-1

注意硬件dev的匹配，如果没有使用lane0，则dev不为0，dev则可以和任意

pipe绑定。

```
sc4336p_4M30.ini  gc4663_4M30.ini
85 stats_cfg.col = 2
86 stats_cfg.interval = 10
87 [mipi]
88 lane_divide_mode = 0
89 ;LANE_DIVIDE_MODE_0 = 0, /* 4lane */
90 ;LANE_DIVIDE_MODE_1 = 1, /* 2lane + 2lane */
91 mipi_mode_num = 2
92 [mipi_mode.0]
93 sns_clk_src = 0
94 sns_rst_src = 0
95 ;ext_data_type_t
96 ext_data_en = TRUE
97 num = 3
98 ext_data_bit_width = 12|12|12
99 ext_data_type = 0x2c|0x2c|0x2c
100 ;combo_dev_attr_t
101 input_mode = 0
102 ;INPUT_MODE_MIPI = 0x0, /* mipi */
103 ;INPUT_MODE_SUBLVDS = 0x1, /* SUB_LVDS */
104 ;INPUT_MODE_LVDS = 0x2, /* LVDS */
105 ;INPUT_MODE_HISPI = 0x3, /* HISPI */
106 img_rect.x = 0 ;oringnal sensor input image size x
107 img_rect.y = 0 ;oringnal sensor input image size y
108 img_rect.width = 2560 ;oringnal sensor input image size W
109 img_rect.height = 1440 ;oringnal sensor input image size H
110 data_rate = 0 ;MIPI_DATA_RATE_X1 = 0
111 ;MIPI_DATA_RATE_X2 = 1
112 ;combo_dev_attr_t->mipi_dev_attr_t
113 mipi_attr.input_data_type = 2
114 ;DATA_TYPE_RAW_8BIT = 0
115 ;DATA_TYPE_RAW_10BIT = 1
116 ;DATA_TYPE_RAW_12BIT = 2
117 ;DATA_TYPE_RAW_14BIT = 3
118 ;DATA_TYPE_RAW_16BIT = 4
119 ;DATA_TYPE_YUV420_8BIT_NORMAL = 5
120 ;DATA_TYPE_YUV420_8BIT_LEGACY = 6
121 ;DATA_TYPE_YUV422_8BIT = 7
122 ;DATA_TYPE_YUV422_PACKED=8
123 mipi_attr.wdr_mode = 0
124 ;OT_MIPI_WDR_MODE_NONE =0
125 ;OT_MIPI_WDR_MODE_VC = 1
126 ;OT_MIPI_WDR_MODE_DT = 2
127 ;OT_MIPI_WDR_MODE_DOL =3
128 mipi_attr.lane_id = 0|2|-1|-1|-1|-1|-1|-1
129 ;lane_id: -1 - disable
130 ;combo_dev_attr_t->lvds_dev_attr_t
131 lvds_attr.input_data_type = 1
132 ;DATA_TYPE_RAW_8BIT = 0
133 ;DATA_TYPE_RAW_10BIT = 1
134 ;DATA_TYPE_RAW_12BIT = 2
135 ;DATA_TYPE_RAW_14BIT = 3
136 ;DATA_TYPE_RAW_16BIT = 4
137 ;DATA_TYPE_YUV420_8BIT_NORMAL = 5
138 ;DATA_TYPE_YUV420_8BIT_LEGACY = 6
139 ;DATA_TYPE_YUV422_8BIT = 7
140 ;DATA_TYPE_YUV422_PACKED=8
141 lvds_attr.wdr_mode = 0
142 ;OT_LVDS_WDR_MODE_NONE =0
143 ;OT_LVDS_WDR_MODE_2F = 1
144 ;OT_LVDS_WDR_MODE_3F = 2
145 ;OT_LVDS_WDR_MODE_4F =3
146 ;OT_LVDS_WDR_MODE_DOL_2F=4
147 ;OT_LVDS_WDR_MODE_DOL_3F=5
148 ;OT_LVDS_WDR_MODE_DOL_4F=6
149 lvds_attr.svnc_mode = 1
```

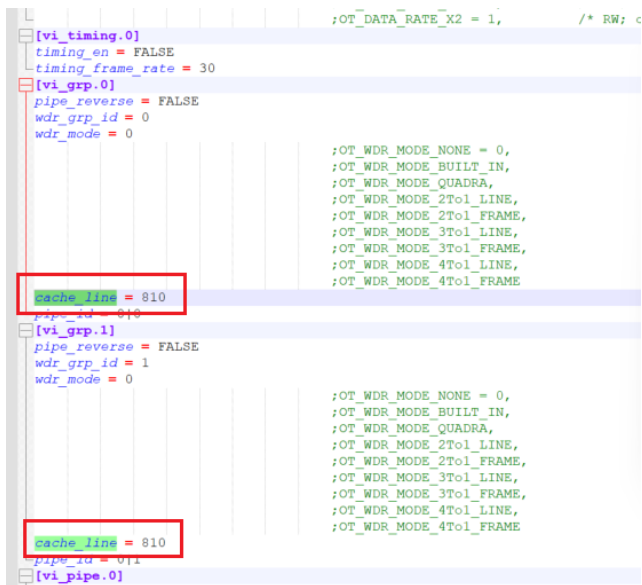
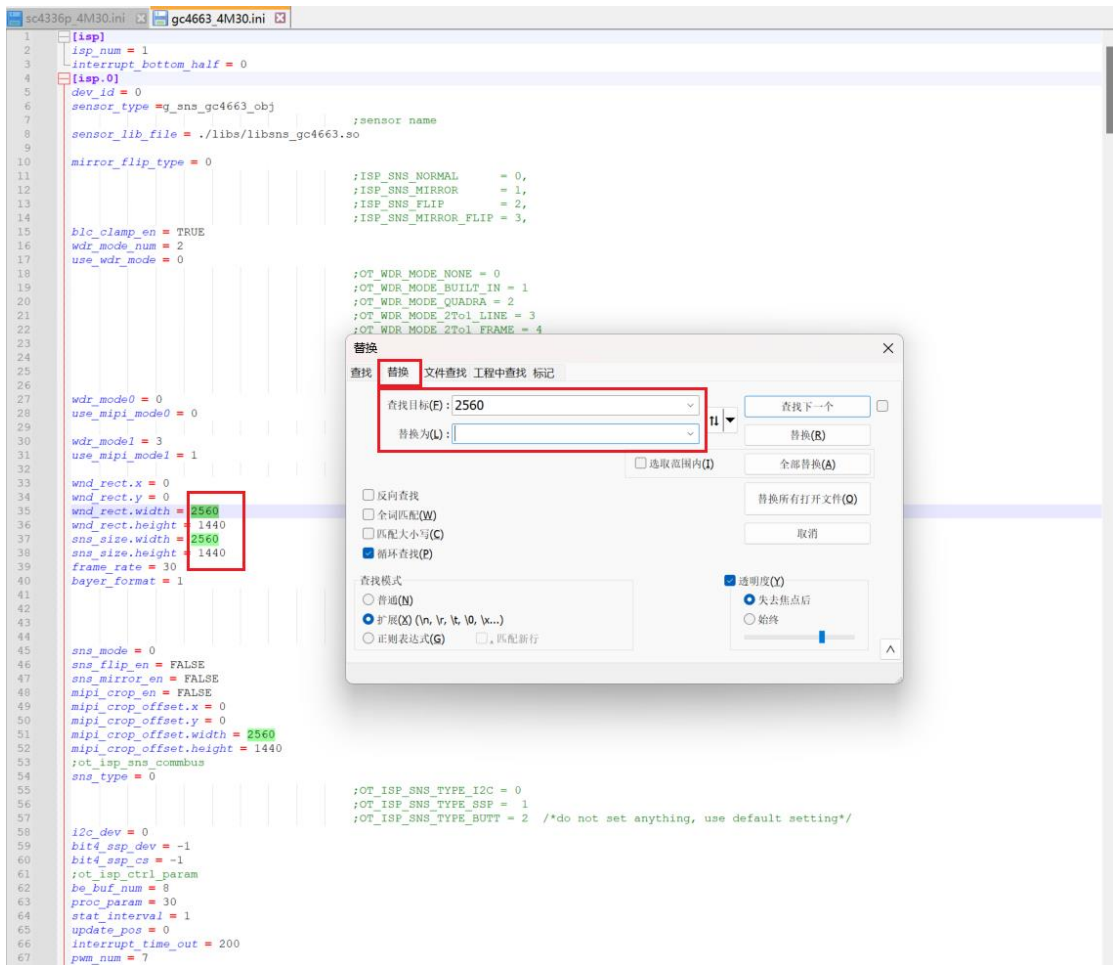
```

sc4336p_4M30.ini  gc4663_4M30.ini
184 lvds_attr.sync_code7 = 0xab0|0xb60|0x800|0x9d0|0xab0|0xb60|0x800|0x9d0|0xab0|0xb60|0x800|0x9d0|0xab0|0xb6
185 lvds_attr.sync_code8 = 0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|
186 lvds_attr.sync_code9 = 0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|
187 lvds_attr.sync_code10 = 0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|
188 lvds_attr.sync_code11 = 0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|
189 [mipi_mode.1]
190 ;
191 sns_rst_src = 0
192 ;ext_data_type_t
193 ext_data_en = TRUE
194 num = 3
195 ext_data_bit_width = 12|12|12
196 ext_data_type = 0x2c|0x2c|0x2c
197 ;combo_dev_attr_t
198 input_mode = 0
199
200 ;INPUT_MODE_MIPI = 0x0, /* mipi */
201 ;INPUT_MODE_SUBLVDS = 0x1, /* SUB LVDS */
202 ;INPUT_MODE_LVDS = 0x2, /* LVDS */
203 ;INPUT_MODE_HISPI = 0x3, /* HISPI */
204
205 ;original sensor input image size x
206 ;original sensor input image size y
207 ;original sensor input image size W
208 ;original sensor input image size H
209
210 data_rate = 0
211 ;MIPI_DATA_RATE_X1 = 0
212 ;MIPI_DATA_RATE_X2 = 1
213
214 ;combo_dev_attr_t->mipi_dev_attr_t
215 mipi_attr.input_data_type = 1
216
217 ;DATA_TYPE_RAW_8BIT = 0
218 ;DATA_TYPE_RAW_10BIT = 1
219 ;DATA_TYPE_RAW_12BIT = 2
220 ;DATA_TYPE_RAW_14BIT = 3
221 ;DATA_TYPE_RAW_16BIT = 4
222 ;DATA_TYPE_YUV420_8BIT_NORMAL = 5
223 ;DATA_TYPE_YUV420_8BIT_LEGACY = 6
224 ;DATA_TYPE_YUV422_8BIT = 7
225 ;DATA_TYPE_YUV422_PACKED=8
226
227 mipi_attr.wdr_mode = 0
228 ;OT_MIPI_WDR_MODE_NONE =0
229 ;OT_MIPI_WDR_MODE_VC = 1
230 ;OT_MIPI_WDR_MODE_DT = 2
231 ;OT_MIPI_WDR_MODE_DOL =3
232
233 mipi_attr.lane_id = 0|2|-1|-1|-1|-1|-1|-1|
234 ;lane_id: -1 - disable
235 ;combo_dev_attr_t->lvds_dev_attr_t
236 lvds_attr.input_data_type = 2
237
238 ;DATA_TYPE_RAW_8BIT = 0
239 ;DATA_TYPE_RAW_10BIT = 1
240 ;DATA_TYPE_RAW_12BIT = 2
241 ;DATA_TYPE_RAW_14BIT = 3
242 ;DATA_TYPE_RAW_16BIT = 4
243 ;DATA_TYPE_YUV420_8BIT_NORMAL = 5
244 ;DATA_TYPE_YUV420_8BIT_LEGACY = 6
245 ;DATA_TYPE_YUV422_8BIT = 7
246 ;DATA_TYPE_YUV422_PACKED=8
247
248 lvds_attr.wdr_mode = 0
249 ;OT_LVDS_WDR_MODE_NONE =0
250 ;OT_LVDS_WDR_MODE_2F = 1
251 ;OT_LVDS_WDR_MODE_3F = 2
252 ;OT_LVDS_WDR_MODE_4F =3
253 ;OT_LVDS_WDR_MODE_DOL_2F=4
254 ;OT_LVDS_WDR_MODE_DOL_3F=5

```

(7) 修改分辨率

为了而避免修改宽高信息改漏，**全局搜索宽高**，一般进行替换和修改。注意vi中的cache_line参数不超过高



(8) 修改vb的大小，如果有vb报错的信息，可以修改这个参数进行vb数量的调整。

```
sc4336p_4M30.ini  gc4663_4M30.ini
770 smart_p.bg_interval = 1200
771 smart_p.bg_qp_delta = 7
772 smart_p.vi_qp_delta = 2
773 ;ot_venc_gop_attr->ot_venc_gop_smart_crr
774 smart_crr.bg_interval = 1200
775 smart_crr.bg_qp_delta = 7
776 smart_crr.vi_qp_delta = 2
777 smart_crr.crr_split_num = 3
778 smart_crr.crr_delay_num = 0
779 smart_crr.strategy = 0
780
781 ;OT_VENC_CRR_RECODE_DISABLE = 0, /* not recode frame */
782 ;OT_VENC_CRR_RECODE_NORMAL = 1, /* recode frame for display */
783 ;OT_VENC_CRR_RECODE_COPY = 2, /* copy next frame stream to replace
784 ;OT_VENC_CRR_RECODE_WRITE_BACK = 3, /* multi chn use one write back
785
786 [bind]
787 bind_num = 2
788 [bind_attr]
789 ;SrcMod|SrcDev|SrcChn|DstMod|DstDev|DstChn
790 bind0 = 16|0|0|7|0|0
791 bind1 = 7|0|0|8|0|0
792 ;SrcMod and DstMod support list
793 ;OT_ID_VPSS = 7
794 ;OT_ID_VENC = 8
795 ;OT_ID_VI = 16
796
797 [vb]
798 vb_num = 1
799 supplement_cfg = 0x8
800
801 #define OT_VB_SUPPLEMENT_JPEG_MASK 0x1
802 #define OT_VB_SUPPLEMENT_MOTION_DATA_MASK 0x2
803 #define OT_VB_SUPPLEMENT_DNG_MASK 0x4
804 #define OT_VB_SUPPLEMENT_BNR_MOT_MASK 0x8
805
806 [vbbk.0]
807 vb_size = 2560|1440
808 ;width|height
809 vb_coefficient = 15
810 ;blk_size = width*height*vb_coefficient/10
811 blk_cnt = 8
812
813 [vbbk.1]
814 vb_size = 2560|1440
815 ;width|height
816 vb_coefficient = 20
817 ;blk_size = width*height*vb_coefficient/10
818 blk_cnt = 2
819
820 [sys]
821 align = 64
822 vi_aisp_mode = 0
823
824 ;OT_VI_AISP_MODE_DEFAULT = 0,
825 ;OT_VI_AISP_MODE_AIDRC = 1,
826 ;OT_VI_AISP_MODE_AIDM = 2,
827 ;OT_VI_AISP_MODE_AI3DNR = 3,
828
829 vi_vpss_mode = 1|1|1|1|0|0|0|0|0|0|0|0|0|0|0|0|0|0
830 ;OT_VI_OFFLINE_VPSS_OFFLINE = 0,
831 ;OT_VI_OFFLINE_VPSS_ONLINE = 1,
832 ;OT_VI_ONLINE_VPSS_OFFLINE = 2,
833 ;OT_VI_ONLINE_VPSS_ONLINE = 3,
834 ;OT_VI_PARALLEL_VPSS_OFFLINE = 4,
835 ;OT_VI_PARALLEL_VPSS_ONLINE = 5
836
837 comp_ratio_10_bit = 2000
838 comp_ratio_12_bit = 2000
839 comp_ratio_14_bit = 2000
840
```

(8) 调整设置vi vpss离线

将vi_vpss_mode设置为全0，因为在vi离线模式下，vi中能显示帧率，方便调试。vi_vpss_mode = 0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0

